

The empirical recurrence for OEIS A254392, recovered from its tail

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Abstract

OEIS A254392 records “Empirical recurrence of order 68” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 174$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 71$. Its last coefficient is $c_{68} = -1231171548132409344 \neq 0$, so no further tail satisfies anything shorter; any order-68 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A254392 is “Number of $(3+2)X(n+2)$ 0..1 arrays with every $3X3$ subblock diagonal maximum plus antidiagonal maximum nondecreasing horizontally, vertically and ne-to-sw antidiagonally.”. It has offset 1 and begins

21304, 409051, 8533796, 174326376, 3575408117, 73650997253, 1515512093870, 31188009589764, ...

The entry states:

Empirical recurrence of order 68 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 14:38:59, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 174$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 174$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 68: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 68.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 348$ exact terms from $n = 3$ on, it returns order 68 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{68} c_i a(n-i),$$

c_1	28	c_{18}	32699882935471	c_{35}	21088569848554387200	c_{52}	−275833828457276
c_2	−145	c_{19}	−80824418372722	c_{36}	−6175948085185980672	c_{53}	−473425081035810
c_3	−290	c_{20}	56284941969440	c_{37}	−35119055493109299200	c_{54}	1186363267494275
c_4	3956	c_{21}	−46956907765996	c_{38}	87287433883317381120	c_{55}	−1092181383697229
c_5	−34092	c_{22}	−1251829135370140	c_{39}	−71152746818372299776	c_{56}	3550825316851541
c_6	509816	c_{23}	7345279239088264	c_{40}	−1193184180774223872	c_{57}	2164984478479417
c_7	−9183232	c_{24}	−15642954340140224	c_{41}	79838548171995246592	c_{58}	−597837456528830
c_8	47960426	c_{25}	34991707665592528	c_{42}	−145197454741129986048	c_{59}	5526564159110051
c_9	37779032	c_{26}	−30230892226810528	c_{43}	98219181166194999296	c_{60}	−175101205924628
c_{10}	−678709222	c_{27}	−167431433226489088	c_{44}	46278324649743876096	c_{61}	−382039107756946
c_{11}	2723203204	c_{28}	525110680374564432	c_{45}	−234303966570154885120	c_{62}	1100951823081642
c_{12}	−16659134876	c_{29}	−858915930555997632	c_{46}	414271824050006786048	c_{63}	−101657551517087
c_{13}	53539215576	c_{30}	942642794728024464	c_{47}	−338898926865739415552	c_{64}	3011881646655170
c_{14}	−113330978436	c_{31}	−581448516624828640	c_{48}	28683437128719269888	c_{65}	269960011335283
c_{15}	67284010856	c_{32}	−1137297384843599680	c_{49}	472882511558017548288	c_{66}	−46938415272548
c_{16}	1656306886479	c_{33}	9956885340665483712	c_{50}	−1029184145759225774080	c_{67}	410390516044136
c_{17}	−8030982358532	c_{34}	−22016001838561733760	c_{51}	928564152237266305024	c_{68}	−12311715481324

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 71$, and it is the recurrence OEIS A254392 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{68} - c_1 x^{68-1} - \dots - c_{68}$. Its constant term is $-c_{68} = 1231171548132409344$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the

components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 68 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 68, so $\deg q = 68$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 68, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 14 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 68, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -1231171548132409344 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-68 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A254392.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.